Instructions for *ACL Proceedings

Anonymous ACL submission

Abstract

foundational algorithms (Aho and Ullman, 1972). This is a placeholder

1 Introduction

2 Results

Language	DeBERTa	Qwen	XLM-RoBERTa
Acholi	0.1994	0.0989	0.0438
Luganda	0.1933	0.1253	0.0853
Masaaba	0.1800	0.0817	0.0814
Nyankole	0.1924	0.1210	0.0865
Soga	0.1911	0.1170	0.0946
Akan	0.1891	0.1277	0.0651
Ewe	0.1813	0.0633	0.0527
Dagbani	0.2281	0.0730	0.0509
Norwegian	0.1251	0.0663	0.0744
Slovak	0.1232	0.0656	0.0751
Serbian	0.1313	0.1568	0.1456
Bulgarian	0.1145	0.0710	0.0636
Czech	0.1216	0.0634	0.0741
Turkish	0.1389	0.0700	0.0819
Persian	0.1374	0.0818	0.0657
Vietnamese	0.1361	0.0614	0.0720
Indonesian	0.0958	0.0566	0.0727
Assamese	0.1815	0.0419	0.0648
Dogri	0.1576	0.0957	0.0506
GOM	0.1757	0.0730	0.0554
Kannada	0.1461	0.0429	0.1174
Nepali	0.1900	0.0551	0.0296
Tamil	0.1330	0.0468	0.0689
Sindhi	0.1466	0.0534	0.0461

Table 1: Silhouette Scores per Language Across Models

References

Alfred V. Aho and Jeffrey D. Ullman. 1972. *The Theory of Parsing, Translation and Compiling*, volume 1. Prentice-Hall, Englewood Cliffs, NJ.

Language 1	Language 2	Matrix Shape	Mean Cosine Sim.
Acholi	Luganda	(4108, 5455)	0.9508
Acholi	Masaaba	(4108, 6867)	0.9619
Acholi	Nyankole	(4108, 6783)	0.9583
Acholi	Soga	(4108, 6219)	0.9615
Acholi	Akan	(4108, 10107)	0.9490
Acholi	Ewe	(4108, 15054)	0.9513
Acholi	Dagbani	(4108, 14231)	0.9727
Acholi	Norwegian	(4108, 534)	0.6627
Acholi	Slovak	(4108, 538)	0.6109
Acholi	Serbian	(4108, 546)	0.6337
Acholi	Bulgarian	(4108, 539)	0.5994
Acholi	Czech	(4108, 529)	0.5950
Acholi	Turkish	(4108, 535)	0.6592
Acholi	Persian	(4108, 535)	0.7178
Acholi	Vietnamese	(4108, 530)	0.4640
Acholi	Indonesian	(4108, 548)	0.6435
Acholi	Assamese	(4108, 4829)	0.7712
Acholi	Dogri	(4108, 4874)	0.8890
Acholi	GOM	(4108, 4831)	0.9279
Acholi	Kannada	(4108, 11643)	0.7280
Acholi	Nepali	(4108, 4814)	0.7242
Acholi	Tamil	(4108, 12841)	0.7609
Acholi	Sindhi	(4108, 4831)	0.7811
Luganda	Nyankole	(5455, 6783)	0.9677
Luganda	Soga	(5455, 6219)	0.9679
Norwegian	Slovak	(534, 538)	0.9842
Norwegian	Serbian	(534, 546)	0.9835
Slovak	Czech	(538, 529)	0.9935
Kannada	Nepali	(11643, 4814)	0.9748
Nepali	Sindhi	(4814, 4831)	0.9694

Table 2: Cross-Lingual Cosine Similarity Metrics (DeBERTa Model)

Language 1	Language 2	Matrix Shape	Mean Cosine Sim.
Acholi	Luganda	(4108, 5455)	0.9033
Acholi	Masaaba	(4108, 6867)	0.8999
Acholi	Nyankole	(4108, 6783)	0.9018
Acholi	Soga	(4108, 6219)	0.9156
Acholi	Akan	(4108, 10107)	0.8560
Acholi	Norwegian	(4108, 534)	0.5413
Acholi	Bulgarian	(4108, 539)	-0.1191
Acholi	Persian	(4108, 535)	-0.1207
Acholi	Kannada	(4108, 11643)	-0.0689
Luganda	Nyankole	(5455, 6783)	0.9606
Norwegian	Slovak	(534, 538)	0.4964
Slovak	Czech	(538, 529)	0.8998
Dogri	GOM	(4874, 4831)	0.9543
Kannada	Tamil	(11643, 12841)	0.5917

Table 3: Cross-Lingual Cosine Similarity Metrics (Qwen Model)

Language 1	Language 2	Matrix Shape	Mean Cosine Sim.
Acholi	Luganda	(4108, 5455)	0.9965
Acholi	Masaaba	(4108, 6867)	0.9962
Acholi	Nyankole	(4108, 6783)	0.9967
Acholi	Norwegian	(4108, 534)	0.9894
Luganda	Nyankole	(5455, 6783)	0.9971
Norwegian	Slovak	(534, 538)	0.9974
Slovak	Czech	(538, 529)	0.9987
Vietnamese	Indonesian	(530, 548)	0.9983
Dogri	GOM	(4874, 4831)	0.9980
Kannada	Tamil	(11643, 12841)	0.9975

Table 4: Cross-Lingual Cosine Similarity Metrics (XLM-RoBERTa Model)

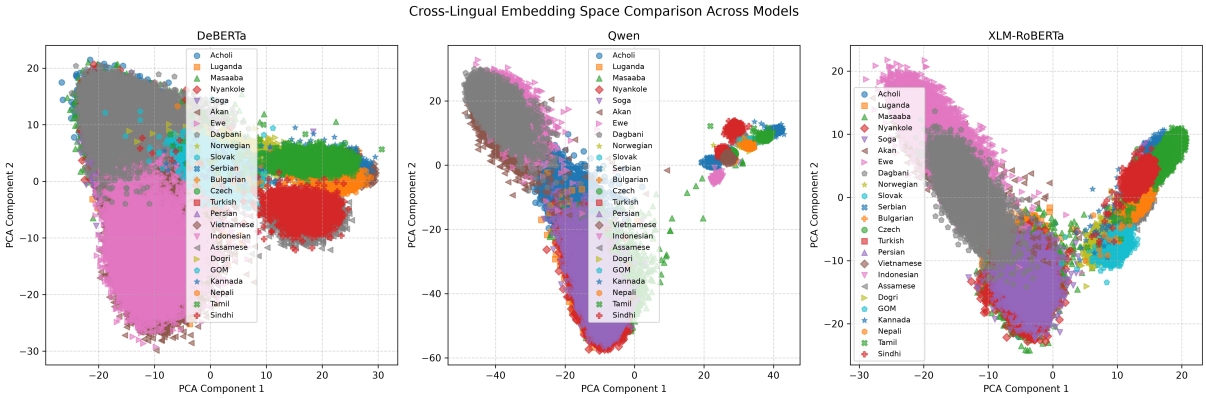


Figure 1: A minimal working example to demonstrate how to place two images side-by-side.